# ConvPE: Convolutional Position Embedding Improves Transformers

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## ABSTRACT

ConvPE (Convolutional Position Embedding) improves Transformer performance and efficiency across language modeling, long-context extrapolation, and vision. On 3B tokens of FineWeb-Edu-10BT, it reduces final training perplexity by 8.41% and raises the mean of eight zero-shot accuracy metrics by 2.35% relative to a RoPE baseline, while using only 2.64% more parameters and 2.71% more forward-pass time. It matches the baseline's final loss with 31.3% fewer training tokens and about 27% less wall-clock training time. ConvPE also extrapolates better: on 1M training tokens it reduces perplexity by 71.1% at 4K and 89.8% at 8K contexts, and on RULER it raises average accuracy from 16.3% to 21.1% at 1K and from 3.0% to 12.3% at 2K. In vision, ConvPE2D improves CIFAR-100 accuracy by 10.9% relative to RoPE. ConvPE realizes these gains with a windowed, content-dependent attention-score bias generated by a lightweight causal convolution and added directly to the scores. Because the bias is added without altering the queries or keys, ConvPE composes with RoPE, arbitrary masks, KV caching, and n-dimensional convolutions, and leaves positions outside the window unbiased. The code is publicly available at `https://github.com/wzhagi/convpe`.

***Keywords*** Position Embedding · Language Models · Transformers · Attention · Vision Transformers

## 1 Introduction

Transformers[1] are insensitive to input order, whereas word order is crucial for language understanding and the spatial layout of image patches matters for vision. Explicit or implicit position embeddings are therefore needed to inject order information into the model[2,3,4]. Where this information is injected affects not only accuracy but also what the model can do at test time, especially when the context is longer than the training length.

Position information is currently injected at one of three points. Absolute and learnable embeddings add a position vector to the token representations[1,2,3]; they are simple, but every position must be represented, and a learned table does not transfer to inputs longer than those seen in training. Rotary position embedding (RoPE)[4] rotates queries and keys so that attention scores depend on relative distance; it is the default in many language models, and extending its context requires frequency rescaling or interpolation[5]. Attention-score-bias methods add a term directly to the score matrix: ALiBi[6] uses a fixed linear decay, CABLE[7] learns to modulate a fixed distance-penalty skeleton, Composite[8] adds convolution-based relative-position terms, and DAPE[9] convolves the full score map, which is effective but adds another pass over a matrix that grows with the square of the context length.

These families thus couple the positional signal to the input representation, to the queries and keys, or to a distance prior

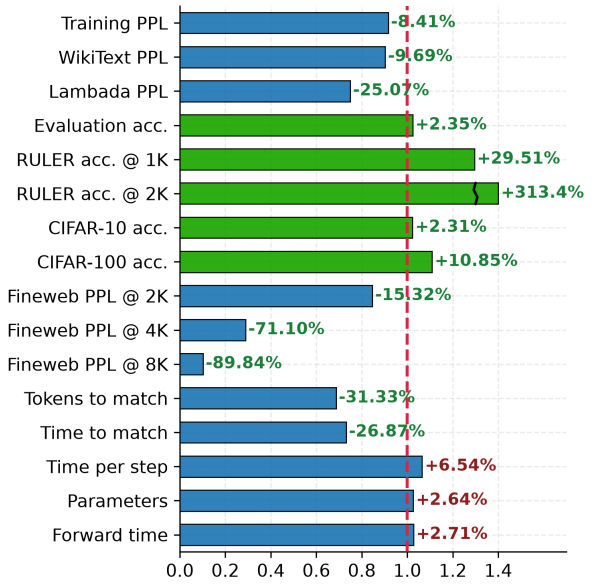



Figure 1: ConvPE relative to the RoPE baseline (dashed line at 1.0). Blue bars are lower-is-better metrics; green bars are higher-is-better metrics.

on the scores. This paper takes a fourth route: leave the queries and keys unchanged, and generate a small positional bias locally. ConvPE projects each hidden state to $C$ channels and applies a causal convolution with window $W$ to produce $H_b \times W$ biases for the current query. The bias for relative offset $j$ is added to the score of the key $j$ positions back, for $j < W$; beyond the window, the score is unchanged. The module is content-dependent because its input is the current hidden state, and it is bounded because the convolution only sees the last $W$ tokens. Its cost is one $D \times C$ projection and one $W \times C \to W \times H_b$ convolution per layer. Because the bias is separate from $Q$ and $K$, ConvPE can be combined with RoPE, used with any attention mask, evaluated with a KV cache, and extended to two- or three-dimensional inputs by replacing the 1D convolution with a 2D or 3D one.

We evaluate ConvPE on language-model pretraining with 3B tokens of FineWeb-Edu-10BT, zero-shot evaluation on nine language tasks and twelve RULER tasks, and image classification with ViT-Tiny on CIFAR-10 and CIFAR-100. Figure 1 summarizes the relative changes with respect to the RoPE baseline: RULER and evaluation accuracies are ratios of the two models' average accuracies, and the token/time rows give the cost of reaching the baseline's final loss. Tables 1 to 6 report the underlying numbers.

**Performance.** ConvPE lowers the final training perplexity by 8.41% and raises the mean of eight zero-shot accuracy metrics by 2.35% relative to the RoPE baseline. On vision, ConvPE2D improves CIFAR-100 accuracy by 10.85% relative (69.5% vs 62.7%).

**Training and token efficiency.** ConvPE adds 2.64% parameters and 2.71% forward-pass time, and a training step becomes 6.5% slower. It reaches the baseline's final loss with 31.3% fewer training tokens, so the total wall-clock time needed to reach that loss falls by about 27%.

**Context extension.** On 1M training tokens, ConvPE reduces perplexity by 8.9% at 1K, 15.3% at 2K, 71.1% at 4K, and 89.8% at 8K contexts. On RULER, mean accuracy rises from 16.3% to 21.1% at 1K and from 3.0% to 12.3% at 2K.

**The main contributions of this paper are as follows:**

- We propose ConvPE, a convolutional attention-score-bias position embedding that combines strong performance with broad compatibility.
- We study the influence of ConvPE hyperparameters and architecture choices, identifying a simple default configuration with a favorable performance-efficiency balance.
- We demonstrate that ConvPE improves training efficiency, token efficiency, and zero-shot long-context extrapolation in language models.
- We verify the effectiveness of ConvPE2D in vision transformers on CIFAR-10 and CIFAR-100.
- We develop high-performance TileLang [10] kernels for ConvPE.

## 2 Background

### 2.1 Attention and Score Biases

Given the input query $Q \in \mathbb{R}^{S_q \times D_k}$, key $K \in \mathbb{R}^{S_{kv} \times D_k}$, and value $V \in \mathbb{R}^{S_{kv} \times D_v}$, the computation of scaled dot-product attention [1] $O \in \mathbb{R}^{S_q \times D_v}$ is

$$O = \mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{D_k}} + \mathbf{B} + \mathbf{M}\right) V, \tag{1}$$

where $S_q$ is the query sequence length, $S_{kv}$ is the key/value sequence length, $\mathbf{B} \in \mathbb{R}^{S_q \times S_{kv}}$ is an additive attention-score bias, and $\mathbf{M} \in \mathbb{R}^{S_q \times S_{kv}}$ is a mask. A causal model sets $M_{ij} = -\infty$ for $j > i$, so that a query at position $s$ can attend only to keys at positions $t \le s$. With batch size and multi-head settings, $Q, K, V, O$ and the two bias terms carry additional batch and head dimensions; we omit them when they are unambiguous.

### 2.2 Position Embeddings

Existing position embedding schemes can be roughly divided into five categories:

1. **Absolute Position Embedding** [1]: adds a fixed positional vector, usually sine waves, to the input. It is simple to implement, but its performance is usually limited and it is not content-dependent.
2. **Learnable Positional Bias** [2,3]: adds a learnable positional vector to the input sequence. It is typically used when the maximum length is fixed, such as in fixed-resolution images or text truncated to a fixed length.

3. **Rotary Position Embedding**[4]: encodes position as a rotation matrix and applies it to queries and keys, so that attention scores depend on relative positions. RoPE has been widely adopted in modern large language models, and it enables long-context extension through frequency scaling, such as YaRN[5].
4. **Attention Score Bias**: adds a positional bias $\mathbf{B}$ to attention scores[11,6,8]. RoPE can be viewed as a rotation-based equivalent, as it also makes attention scores depend on relative positions.
5. **No Position Embedding (NoPE)**[12]: adds no explicit position embeddings; instead, it relies on the model structure and causal attention masks to learn positional information implicitly. For example, Kimi K3 applies NoPE to its global-attention layers[13].

ConvPE belongs to the fourth category, but differs from existing score-bias methods in two respects. First, the bias is produced by a bounded causal convolution over the hidden states, rather than by a fixed distance function or by a convolution over the full score matrix. Second, the bias is applied only to the $W$ most recent keys; beyond the window, the bias is exactly zero, so the model falls back to content-based attention. Section 3 makes this precise.

## 3 Method

### 3.1 Windowed Convolutional Score Bias

Given token representations $X \in \mathbb{R}^{N\times S\times D}$, ConvPE first projects them to a small channel dimension,

$$X_{\text{down}} = XW_p, \qquad W_p \in \mathbb{R}^{D\times C}, \tag{2}$$

and then applies a causal convolution with window $W$,

$$O_{\text{Conv}} = \text{CausalConv1d}\left(X_{\text{down}};\, W,\, C \to H_b W\right), \tag{3}$$

where $O_{\text{Conv}} \in \mathbb{R}^{N\times S\times H_b W}$ and the convolution is left-padded by $W-1$ positions. The output is reshaped into the score bias $\mathbf{B}_{\text{ConvPE}} \in \mathbb{R}^{N\times S\times H_b\times W}$. For a query at position $s$ and a key at position $t \le s$, the bias added to the attention score of head $h$ is

$$\mathbf{B}_{n,h,s,t} = \begin{cases} \mathbf{B}^{\text{ConvPE}}_{n,s,\,h \bmod H_b,\,s-t}, & 0 \le s-t < W, \\ 0, & \text{otherwise}, \end{cases} \tag{4}$$

where $N$ is the batch size, $H$ is the number of attention heads, and $H_b$ must divide $H$. In words, head $h$ reads its bias from bias head $h \bmod H_b$, so each bias head is shared by $H/H_b$ attention heads. Each bias head therefore learns a different relative-position pattern. Future keys are removed by the causal mask, and keys older than $W$ are scored without a positional bias. Figure 2 illustrates the module and the resulting bias pattern.

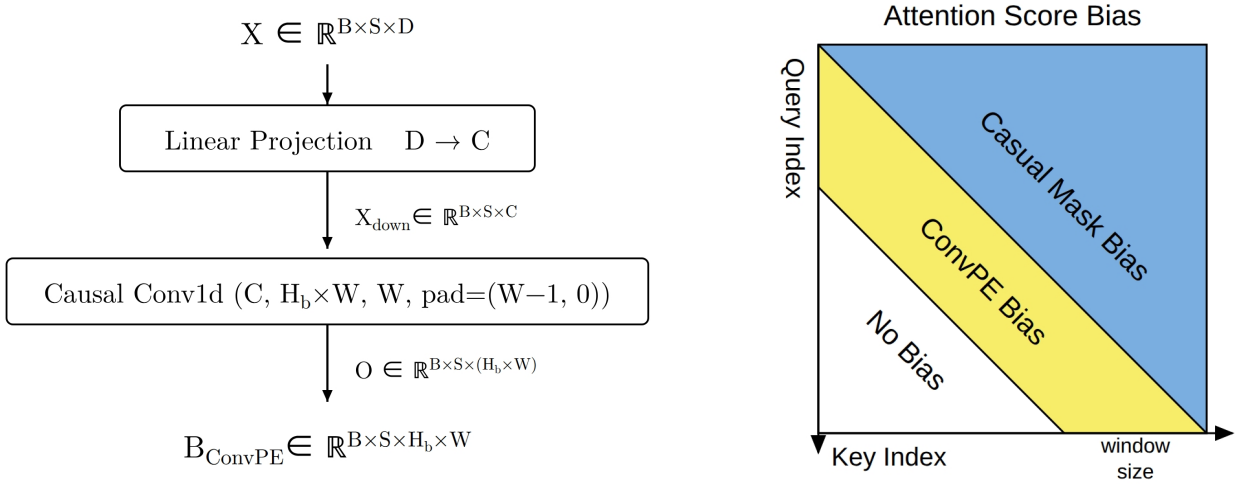



Figure 2: Left: ConvPE projects the hidden states to $C$ channels and applies a causal convolution with window $W$ to produce $H_b \times W$ score biases per position. Right: the bias is added only inside the $W$-token window (yellow); future positions are masked (blue) and visible positions beyond the window receive no bias (white).

The hyperparameters are:

- $W$: convolution window size.
- $C$: output dimension of the linear projection and input channels of the convolution.
- $H_b$: number of score-bias heads.

The projection has $DC$ parameters and the convolution has $CH_bW^2$ parameters. Each parameter is used once per token, so the multiply–accumulate count per token is the same. Together with the $H_bW$ bias terms, a layer has $DC + CH_bW^2 + H_bW$ parameters; for the default configuration this is 3.29M over 12 layers, which accounts for the $+2.64\%$ parameter increase reported in Section 4.3.

### 3.2 Parameter-Efficient Variants

To reduce the convolution parameter count, we introduce three additional hyperparameters:

- $\mathrm{Groups}$: number of convolution groups, which split the $C$ input channels and the $H_bW$ output channels into $G$ groups. Each group has its own kernel, and the parameter count becomes $CH_bW^2/G$.
- $\mathrm{Share}$: whether to share one kernel across all groups. Sharing divides the convolution parameter count by another factor of $G$ (for $G = 1$ it has no effect).
- $\mathrm{Rank}$: rank $R$ of a low-rank factorization of the convolution. A group-shared down-projection maps the $(C/G)W$ input window to $R$ dimensions, and a group-wise up-projection maps $R$ back to the $H_bW$ outputs. This costs $(C/G)WR + RH_bW$ weight parameters instead of $CH_bW^2/G$, plus the same $H_bW$ bias term.

The default configuration uses $G = 1$, no sharing, and $R = 0$ (no factorization).

### 3.3 Properties and Scope

**Compatibility.** ConvPE only adds a bias to the attention scores, so it can be combined with RoPE applied to queries and keys, and with any mask (causal, document, sliding-window, or padding) because the mask is applied separately. For autoregressive decoding, the convolution needs only the projected states of the last $W - 1$ tokens; storing them next to the KV cache costs $C(W - 1)$ values per layer, independent of the context length. Replacing the 1D convolution with an $n$-dimensional convolution and indexing the corresponding neighborhood yields ConvPE2D, ConvPE3D, and so on.

**Scope.** The bias is exactly zero outside the window. ConvPE therefore imposes no positional prior beyond $W$ tokens, and in that region the model behaves like NoPE and must rely on content. This is a deliberate locality assumption: we do not claim that position is irrelevant at long range, only that a learned local bias is sufficient to improve the tasks we study. Section 4.5 tests whether this behavior extrapolates beyond the training length.

## 4 Experiments

### 4.1 Setup

**Architecture.** Unless stated otherwise, all language models have $N_{\text{layers}} = 12$ and use pre-RMSNorm, SwiGLU[14] feed-forward layers with $8 \times D_{\text{model}}^2$ parameters, multi-query attention (MQA)[15] with a single key/value head ($K = V$: one projection feeds both keys and values[16]), query L2 normalization, key RMSNorm, a learnable attention sink per head, and tied input/output embeddings. Each attention layer has $D_{\text{model}} \times D_{\text{head}} \times (2N_{\text{heads}} + 1)$ parameters, excluding normalization scales and the per-head sink. The tiny backbone has $D_{\text{model}} = 192$, $N_{\text{heads}} = 3$, $D_{\text{head}} = 64$, and the base backbone has $D_{\text{model}} = 768$, $N_{\text{heads}} = 12$, $D_{\text{head}} = 128$ (so the query and output projections are 1536-dimensional). The RoPE baseline uses a rotary dimension of 64 and base 10000. The vocabulary has 50,304 entries (the GPT-2 tokenizer[3] padded to a multiple of 128).

**Data.** Pretraining uses the first 3B tokens of FineWeb-Edu-10BT[17], packed into 1024-token blocks. The hyperparameter search trains for one epoch on WikiText-103[18] and reports the held-out loss on the WikiText-2 test split.

**Optimization.** For pretraining, Muon[19] optimizes the hidden 2D parameters at a learning rate of $3 \times 10^{-3}$ with momentum 0.95 and weight decay 0.01, while Adam[20] optimizes the remaining parameters at $3 \times 10^{-4}$. Both use a cosine schedule with 500 warm-up steps. We use a per-device batch size of 4 with 4 gradient-accumulation steps (16 sequences, 16,384 tokens per step), sequence length 1024, BF16 autocast, gradient clipping at 4.0, 183,204 steps (3.0B tokens), and seed 42. The language models are trained with the Hugging Face Trainer[21]. The hyperparameter search uses AdamW[22] at $3 \times 10^{-4}$ with weight decay 0.01, a cosine schedule, per-device batch size 4, sequence length 1024, one epoch, BF16, gradient clipping at 1.0, and seed 42. All language-model evaluations are zero-shot. All

language-model results come from a single run; small task-level differences should be read with sampling noise in mind.

### 4.2 Hyperparameter Search

We train 10 tiny and 8 base configurations and select by held-out loss after one epoch. Model names record the backbone (`tiny` or `base`), the position mechanism (`rope` or `nope`), and the ConvPE fields $W$, $C$, $R$, $H_b$, and $G$ as `w`, `d`, `r`, `h`, and `g` followed by their values, with a trailing `s` when the convolution kernel is shared across groups. For example, `tiny rope w128d6r128h3g1s` denotes a tiny backbone with RoPE, $W = 128$, $C = 6$, $R = 128$, $H_b = 3$, $G = 1$, and weight sharing. Fields equal to the default (position mechanism `nope`, $R = 0$, $H_b = 1$, $G = 1$, and no sharing) are omitted when this is unambiguous, e.g., `base w128d16` is `base nope w128d16r0h1g1`.

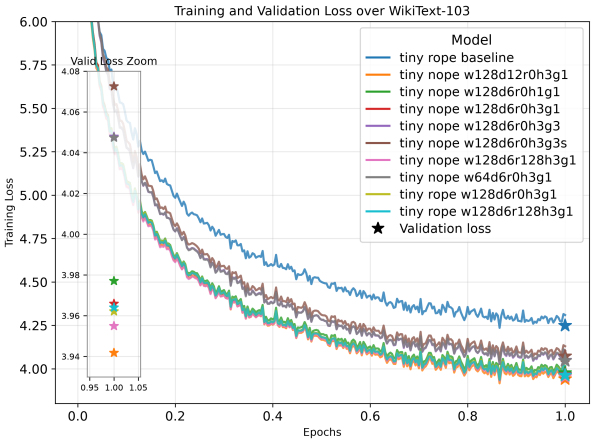



Figure 3: Tiny models on WikiText-103. Curves are training loss; stars mark the held-out WikiText-2 loss used for selection.

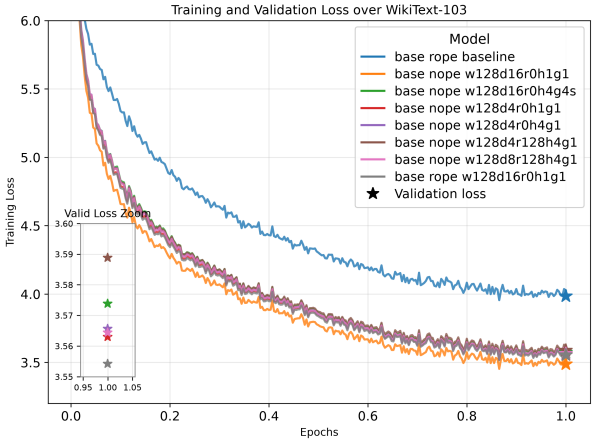



Figure 4: Base models on WikiText-103, with the same conventions as Figure 3.

First, we train tiny models around `nope w128d6r0h3g1`. Figure 3 shows that every ConvPE variant outperforms the RoPE baseline (4.249 vs 3.942–4.073). Increasing the projection dimension from 6 to 12 (3.966 $\rightarrow$ 3.942) and the window from 64 to 128 (4.047 $\rightarrow$ 3.966) give the largest gains. Grouped convolution hurts the most (4.048 for `g3` and 4.073 for `g3s`), and a smaller window costs about as much (4.047). The rank-128 variant (3.955), RoPE integration (3.962), and the full kernel (3.966) are close, whereas $H_b = 1$ is 0.011 worse in this sweep (3.977).

We then train base models around `nope w128d4r0h4g1` and `nope w128d16r0h1g1`. Figure 4 shows that every ConvPE variant again beats the RoPE baseline (3.986). The best configuration is `w128d16r0h1g1` (3.489), followed by the RoPE-integrated variant (3.554), `w128d4r0h1g1` (3.563), `w128d8r128h4` (3.564), `w128d4r0h4` (3.566), `w128d16g4s` (3.574), and `w128d4r128h4` (3.589). Because $H_b = 1$ is also the cheapest bias-head setting, we adopt `w128d16r0h1g1` as the default.

We also tried feeding $Q$ (as in Composite[8]), $KV$, or $X$ directly to the convolution; all three performed worse than first projecting $X$ to the low-dimensional $X_{\text{down}}$.

### 4.3 Pretraining

We train the RoPE baseline and ConvPE `w128d16` from scratch on the first 3B tokens of FineWeb-Edu-10BT with the recipe in Section 4.1. Figure 5 shows that ConvPE converges faster and reaches a lower final loss. Averaged over the last 0.1B tokens, the final training loss drops from 3.1195 to 3.0316 ($-2.82\%$ in loss, corresponding to perplexity 22.64 $\rightarrow$ 20.73, or $-8.41\%$). Interpolating the smoothed training curves, ConvPE reaches the baseline's final loss after 2.06B tokens instead of 3B (31.3% fewer tokens).

To measure cost, we implement high-performance Flash MQA kernels in TileLang and benchmark the two configurations on one GPU with batch size 4 and sequence length 1024, averaged over 1,000 iterations after 1,000 warm-up iterations. ConvPE has 128.13M parameters versus 124.84M ($+2.64\%$), and its forward pass takes 23.14 ms versus 22.53 ms ($+2.71\%$). During training, ConvPE processes 2.60 iterations/s versus 2.77 (6.5% more time per step). The per-step overhead is larger than the forward-pass overhead, which is consistent with the ConvPE backward kernel being less

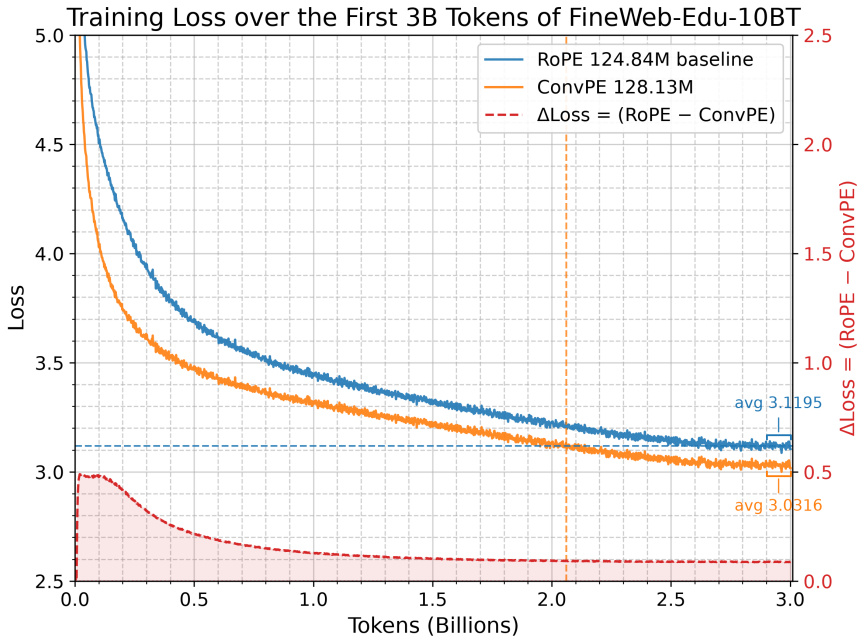



Figure 5: Training loss on the first 3B tokens of FineWeb-Edu-10BT for the RoPE baseline and ConvPE `w128d16`. The dashed curve is the loss difference (RoPE minus ConvPE).

optimized than its forward kernel. Combining the $31.3\%$ token reduction with the $6.5\%$ per-step overhead gives a net reduction of about $27\%$ in the wall-clock time needed to match the baseline's loss. The exact value depends on how the curves are smoothed, so we report it as approximate.

### 4.4 Attention Analysis

To understand the role of the ConvPE module, we analyze the kernel weights and the static output bias of the trained `base w128d16` model. Since the average weights are close to zero, the standard deviation indicates **feature selectivity**: a higher standard deviation means that the kernel focuses on specific channels. Figure 6 shows three properties.

**1. Near-end dominance.** Near the current token, both the static bias and the input-feature selectivity reach their largest magnitudes. This indicates that the positional relationship between nearby tokens is important: ConvPE applies highly specific feature extraction and strong positional priors to the immediate local context.

**2. Far-end relaxation.** Toward the far end of the window, the standard deviation is lower and the static bias is close to zero for most layers. ConvPE therefore does not impose a strong static prior at this range, leaving the decision to the convolution over the dynamic input and to the query–key semantic match.

**3. Self-suppression and self-enhancement.** The static bias has a clear layer-wise pattern at the current position. Layer 0 shows self-enhancement with a bias of $+0.5885$, whereas every deeper layer except L4 (a near-zero $+0.0115$) shows self-suppression with a negative bias. This is consistent with deeper layers shifting attention away from the current token and toward the historical context.

### 4.5 Evaluations

This section evaluates the pretrained ConvPE and RoPE models on downstream and context-extension benchmarks without any fine-tuning.

#### 4.5.1 Zero-Shot Downstream Evaluations

Table 1 shows that ConvPE improves every accuracy metric except TruthfulQA-MC1. WikiText-2 word perplexity drops from $46.56$ to $42.05$ ($-9.7\%$), and LAMBADA perplexity from $118.10$ to $88.49$ ($-25.1\%$), with LAMBADA accuracy rising from $0.2507$ to $0.2604$. The mean of the eight accuracy metrics rises from $43.72\%$ to $44.75\%$, a $2.35\%$ relative improvement. TruthfulQA-MC1 is the exception: both models fall below the $25\%$ random-choice baseline

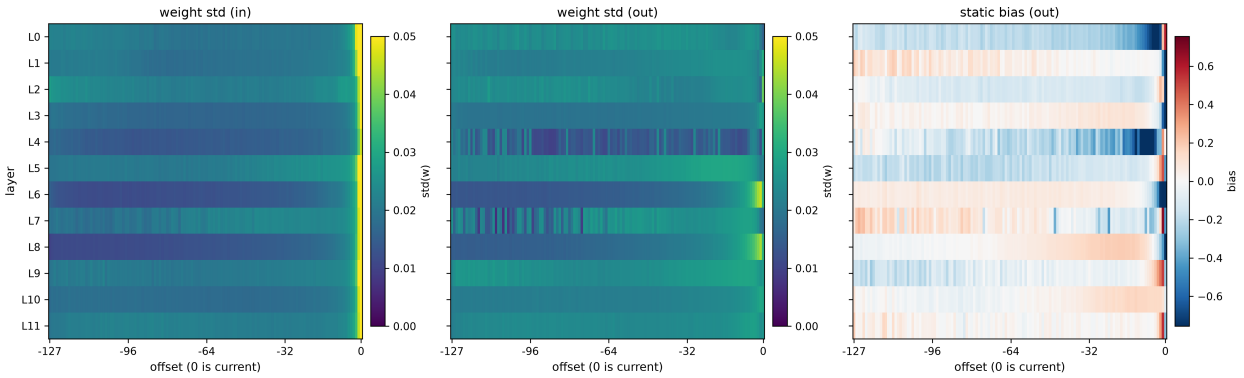



Figure 6: ConvPE statistics of the pretrained `base w128d16` model across layers. Left: standard deviation of the kernel weights in the input dimension. Middle: standard deviation in the output dimension. Right: static output bias, averaged over bias heads. The x-axis is the relative offset ($-127$ to $0$; $0$ is the current token).

| Dataset | Metric | RoPE | ConvPE | Random |
|---|---|---|---|---|
| WikiText-2 | word PPL ↓ | 46.56 | **42.05** | – |
| LAMBADA [23] | PPL ↓ | 118.10 | **88.49** | – |
| LAMBADA | acc ↑ | 0.2507 | **0.2604** | – |
| HellaSwag [24] | acc | 0.2894 | **0.2949** | 0.25 |
| HellaSwag | acc norm | 0.3148 | **0.3272** | 0.25 |
| XStoryCloze [25] | acc | 0.5460 | **0.5526** | 0.5 |
| PIQA [26] | acc | 0.6251 | **0.6338** | 0.5 |
| PIQA | acc norm | 0.6208 | **0.6235** | 0.5 |
| ARC-Easy [27] | acc | 0.5114 | **0.5421** | 0.25 |
| ARC-Easy | acc norm | 0.4487 | **0.4701** | 0.25 |
| WinoGrande [28] | acc | 0.5091 | **0.5193** | 0.5 |
| BoolQ [29] | acc | 0.5676 | **0.5893** | 0.5 |
| TruthfulQA-MC1 [30] | acc | **0.2399** | 0.2375 | 0.25 |
| Average† | acc | 0.4372 | **0.4475** | – |

Table 1: Zero-shot evaluation of the pretrained 125M models on nine language tasks, using lm-evaluation-harness [31]. Lower PPL and higher accuracy are better; the Random column gives the chance level where one exists. †Mean of one accuracy per task (acc-norm for HellaSwag, PIQA, and ARC-Easy, and acc otherwise), excluding the two perplexities.

(23.99% and 23.75%), so this task does not separate them. Overall, the zero-shot results support ConvPE as a stronger position embedding than RoPE for the evaluated language tasks.

### 4.5.2 Context Extension

| | Loss ↓ | | Perplexity ↓ | |
|---|---|---|---|---|
| Context | RoPE | ConvPE | RoPE | ConvPE |
| 1K | 3.1247 | **3.0312** | 22.75 | **20.72** |
| 2K | 3.1684 | **3.0021** | 23.77 | **20.13** |
| 4K | 4.2690 | **3.0278** | 71.45 | **20.65** |
| 8K | 5.6443 | **3.3575** | 282.67 | **28.72** |

Table 2: Language-modeling loss and perplexity on the first 1M tokens of the pretraining corpus, at contexts from 1K to 8K. Both models were trained with 1K-token contexts. These tokens belong to the training split, so the table is a diagnostic of length behavior rather than a held-out generalization test; Table 3 provides held-out evidence.

We first measure language-modeling loss on the first 1M tokens of the pretraining corpus at increasing context lengths. Both models were trained on 1K-token contexts. Table 2 shows a clear advantage for ConvPE as the context grows. RoPE's loss rises sharply beyond 2K (4.269 at 4K and 5.644 at 8K), whereas ConvPE stays between 3.00 and 3.36.

The corresponding perplexity reductions are $8.9\%$ at 1K, $15.3\%$ at 2K, $71.1\%$ at 4K, and $89.8\%$ at 8K. Because these tokens come from the training split, the table measures length behavior rather than generalization; the RULER results below address the latter.

| | 1K | | 2K | | 4K | |
|---|---|---|---|---|---|---|
| Task | RoPE | ConvPE | RoPE | ConvPE | RoPE | ConvPE |
| niah_single_1 | 0.614 | **0.890** | 0.004 | **0.722** | 0.000 | **0.314** |
| niah_single_2 | 0.612 | **0.664** | 0.000 | **0.038** | 0.000 | 0.000 |
| niah_single_3 | 0.090 | **0.180** | 0.000 | **0.014** | 0.000 | 0.000 |
| niah_multikey_1 | 0.198 | **0.200** | 0.118 | **0.194** | 0.000 | **0.002** |
| niah_multikey_2 | **0.002** | 0.000 | 0.000 | 0.000 | – | – |
| niah_multikey_3 | 0.000 | **0.008** | 0.000 | 0.000 | – | – |
| niah_multiquery | 0.104 | **0.186** | 0.029 | **0.179** | 0.000 | 0.000 |
| niah_multivalue | 0.149 | **0.187** | 0.127 | **0.186** | 0.000 | 0.000 |
| ruler_vt | **0.032** | 0.008 | 0.000 | **0.052** | 0.000 | **0.107** |
| ruler_cwe | 0.002 | 0.002 | 0.002 | 0.002 | – | – |
| ruler_fwe | 0.001 | **0.012** | 0.000 | **0.001** | – | – |
| ruler_qa_squad | 0.148 | **0.191** | 0.078 | **0.091** | 0.017 | **0.052** |

Table 3: Accuracy on twelve RULER tasks[32] at three context lengths. Cells marked – were not evaluated at that length. Accuracies are single-run fractions.

We then evaluate both models on RULER[32], which measures long-context retrieval and reasoning on controlled synthetic tasks. Table 3 shows that ConvPE is better on 9 of the 12 tasks at 1K and on 8 of the 12 tasks at 2K; the remaining tasks are tied or, at 1K, lower by less than $0.03$. The mean over the twelve tasks rises from $16.27\%$ to $21.07\%$ at 1K (a $29.5\%$ relative gain) and from $2.98\%$ to $12.33\%$ at 2K ($4.1\times$). At 4K, eight tasks were evaluated: ConvPE retains a mean accuracy of $5.94\%$ versus $0.21\%$ for RoPE, driven by `niah_single_1` ($31.4\%$ vs $0.0\%$), `ruler_vt` ($10.7\%$ vs $0.0\%$), and `ruler_qa_squad` ($5.2\%$ vs $1.7\%$), while both models are near zero on the remaining retrieval tasks. Although ConvPE's language-modeling loss remains stable up to 4K, these downstream results show that zero-shot context extension is still far from solved.

### 4.6 2D Position Embedding

| Model | Params (M) | CIFAR-10 | CIFAR-100 |
|---|---|---|---|
| RoPE baseline (1D) | 5.346 | 0.910 | 0.627 |
| ConvPE2D `w3d4` | 5.359 | 0.931 | **0.695** |
| ConvPE2D `w3d8` | 5.372 | **0.933** | 0.683 |
| ConvPE2D `w5d4` | 5.386 | 0.929 | 0.682 |
| ConvNet (reference) | 4.336 | 0.945 | 0.737 |

Table 4: Test accuracy of ViT-Tiny on CIFAR-10/100 with $4 \times 4$ patches. Bold marks the best position-embedding method per column; ConvNet is a convolutional reference outside the position-embedding comparison. Parameter counts are for the CIFAR-10 models; the CIFAR-100 classifier adds $0.017$M. All models use horizontal-flip test-time augmentation.

For 2D inputs, ConvPE2D replaces the 1D causal convolution with a 2D convolution over an $H \times W$ patch grid: each patch receives a bias over its $w \times w$ neighborhood, and the same window is shared across the grid. We use ViT-Tiny models with multi-head attention (MHA) and the configuration $N_{\text{layers}} = 12$, $D_{\text{model}} = 192$, $N_{\text{heads}} = 3$, and $D_{\text{head}} = 64$, with $4 \times 4$ patches ($64$ tokens). Models are trained on CIFAR-10 and CIFAR-100[33] for $100$ epochs with batch size $256$, using AdamW at $3 \times 10^{-4}$ for the remaining parameters and Muon at $3 \times 10^{-3}$ for the hidden 2D weights. At test time we average the logits of each image and its horizontal flip. The RoPE baseline applies 1D RoPE to the flattened patch sequence; the 2D axial RoPE baseline is considered in Section 4.7.

Table 4 shows that ConvPE2D `w3d4` improves CIFAR-10 and CIFAR-100 accuracy by $2.31\%$ and $10.85\%$ relative to the 1D RoPE baseline, and `w3d8` gives the best CIFAR-10 result ($0.933$). The convolutional reference reaches $0.945$ and $0.737$, so ConvPE2D narrows but does not close the gap between ViT-Tiny and a CNN at this scale. The influence of the 2D hyperparameters requires further study on larger models and datasets.

### 4.7 Comparison with Peer Methods

To situate ConvPE among recent position mechanisms, we use an independent reproduction by Zhijian Qiao (Hong Kong University of Science and Technology), who trained seven methods with the same 12-layer, 768-wide skeleton, the same first 3B tokens of FineWeb-Edu-10BT, 183,204 steps at a global batch of 16, and the same evaluation data. The reproduction leaves the input embedding and the output head untied, so its models have 163–167M parameters, compared with 125–128M in our runs. Absolute values therefore differ from Tables 1 to 3, and comparisons should be made within Table 5. Details are reported in the ConvPE repository (`pe_comparison.md`).

| | | | | FineWeb-Edu 1M PPL ↓ | | | | Eval Acc | RULER Acc | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Method | Params (M) | ms/step | Mem (GiB) | 1K | 2K | 4K | 8K | (%) ↑ | 1K | 2K | 4K |
| NoPE [12] | 163.470 | **141.30** | 10.77 | 24.34 | 173.71 | 1389.25 | 3618.99 | 42.78 | 19.31 | 0.15 | 0.00 |
| Composite [8] | 163.498 | 180.02 | 11.01 | 21.34 | 38.15 | 246.16 | 984.34 | 44.37 | 25.97 | 0.88 | 0.09 |
| RoPE [4] | 163.470 | 149.03 | 10.77 | 22.89 | 24.19 | 55.01 | 207.58 | 43.86 | 11.96 | 2.63 | 0.26 |
| ALiBi [6] | 163.470 | <u>145.67</u> | 10.77 | 21.37 | 20.77 | 20.60 | 20.52 | 43.90 | 14.37 | 3.80 | 1.61 |
| CABLE [7] | 163.691 | 188.05 | 13.17 | 20.71 | 20.01 | <u>19.68</u> | <u>19.52</u> | <u>44.72</u> | 19.80 | 7.69 | 3.22 |
| DAPE V2 [9] | 163.512 | 273.71 | 20.25 | **19.47** | **18.71** | **18.31** | **18.09** | **44.74** | <u>26.11</u> | **17.09** | **7.45** |
| ConvPE | 166.764 | 156.00 | 11.21 | <u>20.52</u> | <u>19.95</u> | 20.67 | 29.33 | 44.66 | **26.58** | <u>14.46</u> | <u>3.49</u> |

Table 5: Comparison with peer position mechanisms in the independent reproduction. PPL is measured on the first 1M pretraining tokens; Eval Acc is the mean of the eight accuracy metrics in Table 1 (acc-norm where available); RULER Acc is the mean over the twelve RULER tasks at each length. Bold and underline mark the best and second-best values per column; parameter counts and memory are not ranked, and memory is peak allocated GPU memory. All rows share the same backbone, data, and evaluation pipeline, but absolute values are not directly comparable to Tables 1 to 3 because the reproduction leaves the embedding and head untied (163–167M parameters vs 125–128M).

Table 5 shows that ConvPE is in the first tier on both the perplexity diagnostic and the accuracy metrics. At 1K it has the second-lowest PPL (20.52) after DAPE V2 (19.47), and at 2K it is again second (19.95 vs 18.71). At 4K and 8K, DAPE V2, CABLE, and ALiBi extrapolate better on this diagnostic: ConvPE reaches 20.67 and 29.33, respectively, while ALiBi stays near 20.5. ConvPE is nonetheless far better than NoPE, Composite, and RoPE at long contexts (e.g., at 4K, 20.67 vs 55.01 for RoPE and 246.16 for Composite; at 8K, 29.33 vs 207.58 for RoPE). On RULER, ConvPE has the best mean accuracy at 1K (26.58%), is second at 2K (14.46%) and 4K (3.49%), and is never below second place. Its downstream accuracy (44.66%) is within 0.08 points of the best result (44.74%). DAPE V2 convolves the full score map, so it is $1.84\times$ slower per step than RoPE (273.71 vs 149.03 ms) and uses 20.25 vs 10.77 GiB of training memory. ALiBi [6] and CABLE [7] are cheaper, but both retain a distance-penalty form. ALiBi applies a fixed linear penalty, and CABLE learns to modulate a distance-penalty skeleton. In both cases the bias contains a component that grows with the offset, so distant keys are penalized; CABLE can adjust the magnitude with content, but the distance component remains. This prior is reasonable for local language modeling, but it conflicts with long-context retrieval, where the relevant evidence may lie far from the query. Prior work supports this concern: Kazemnejad et al. [12] find that ALiBi and other explicit position encodings are not well suited for length generalization on downstream tasks, whereas NoPE, which imposes no distance prior, generalizes better. RULER exhibits the same pattern in our reproduction: ConvPE outperforms both ALiBi and CABLE at every length (26.58% vs 14.37% and 19.80% at 1K; 14.46% vs 3.80% and 7.69% at 2K; 3.49% vs 1.61% and 3.22% at 4K). Because ConvPE applies no bias outside the window, attention beyond $W$ remains content-based, so distant evidence is not suppressed by construction. The trade-off is visible in the 8K language-modeling diagnostic, where ALiBi and CABLE are better (PPL 20.52 and 19.52 vs 29.33); this suggests that a distance penalty can help next-token prediction by smoothing attention even as it hurts retrieval. For long-context tasks that require retrieving distant evidence, we consider the RULER behavior the more relevant one. This comparison covers one scale (163–167M parameters), so we do not claim that the ordering transfers to larger models.

Table 6 reports the CIFAR-100 comparison with 2D adaptations of the same methods. ConvPE2D (67.98%) is the most accurate, 3.81 points above 2D axial RoPE (64.17%) and 1.97 points above DAPE2D V2 (66.01%). The gain over the 1D RoPE baseline in Table 4 (6.8 points) is larger, as expected, because 2D axial RoPE is a stronger baseline. These results indicate that the convolutional bias transfers to grid-structured inputs, although the absolute ranking of 2D position mechanisms remains sensitive to the backbone and training recipe.

| Method | Params (M) | ms/step | CIFAR-100 acc (%) ↑ |
|---|---|---|---|
| NoPE [12] | 5.363 | 66.45 | 56.03 |
| 2D ALiBi [34] | 5.363 | 66.63 | 61.38 |
| Composite2D [8] | 5.596 | 78.29 | 63.50 |
| 2D RoPE [35] | 5.363 | 78.44 | 64.17 |
| Learnable 2D abs. PE [36] | 5.376 | **63.50** | 64.36 |
| Fixed 2D sin-cos [36] | 5.363 | 66.65 | 65.07 |
| CABLE2D [7] | 5.377 | 98.90 | 65.25 |
| DAPE2D V2 [9] | 5.374 | 92.74 | 66.01 |
| ConvPE2D | 5.377 | 82.03 | **67.98** |

Table 6: CIFAR-100 test accuracy in the independent reproduction (ViT-Tiny, 100 epochs). Bold marks the best value per column; parameter counts are not ranked. ConvPE2D is the `w3d4` variant.

## 5 Conclusions and Future Work

ConvPE adds a learned, windowed bias to attention scores without altering the queries or keys. In our experiments it improves pretraining loss, zero-shot accuracy, context extension, and CIFAR accuracy at a $2.6\%$ parameter and $2.7\%$ forward-pass cost. The independent reproduction shows that ConvPE's 8K perplexity is higher than that of ALiBi, CABLE, and DAPE V2, but this does not make those methods better suited to long-context training. ConvPE outperforms ALiBi and CABLE on RULER at every length, and its average downstream accuracy is within $0.08$ points of the best result. DAPE V2 is stronger on the 8K perplexity diagnostic and on RULER at 2K and 4K, but it must materialize and convolve the full score matrix: it is quadratic in context length, incompatible with fused FlashAttention kernels, and therefore impractical for long-context training. ConvPE is the only method among the four that combines competitive downstream accuracy with a FlashAttention-compatible design that has no fixed distance penalty; when accuracy, parameter count, per-step cost, and compatibility are considered jointly, it is the most favorable choice for long-context training.

We adopt $H_b = 1$ without any parameter-saving technique as the default configuration; this configuration achieved the best held-out loss in the base-model sweep and is also the cheapest bias-head setting. ConvPE is compatible with RoPE, various attention masks, and KV-cache concatenation, and it naturally supports multidimensional extensions.

Our study has three main limitations. First, the main experiments use a single seed and models of up to 167M parameters, so the size of the gains at larger scale is unknown. Second, the window is a hard locality assumption: beyond $W$ tokens the model receives no positional bias, which is beneficial for extrapolation but may hurt tasks that require precise long-range positions. Third, the 8K loss diagnostic uses training tokens rather than held-out text. We view these as directions for future work:

- **Kernel optimization.** Further optimize the backward and decode kernels.
- **Scaling behavior.** Investigate ConvPE with larger Transformers and datasets, multiple seeds, and identify configuration patterns at scale.
- **Long-context modeling.** Evaluate ConvPE at contexts of 32K tokens and beyond, and with long-context training rather than zero-shot extension.
- **Architectural integration.** Combine ConvPE with Sliding Window Attention (SWA) [37], Compressed Sparse Attention (CSA) [16], and other Transformer variants.
- **Higher-dimensional modalities.** Extend ConvPE to video, volumetric data, and multimodal inputs with 3D and higher-dimensional convolutions.

## Acknowledgements

We thank Zhijian Qiao (Hong Kong University of Science and Technology) for the independent reproduction and the peer comparison in Section 4.7.

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